Exploratory Data Analysis of Student Performance

Thiranex Internship Project 3

Project report describing the workflow, tools, and analyses performed on the Student Performance dataset.

1. Objective

The objective of this project is to explore student exam-score data, summarize its main characteristics, visualize score distributions, compare groups, and examine relationships between mathematics, reading, and writing scores. The analysis is exploratory and describes associations rather than proving causation.

2. Dataset Overview

The project uses the StudentsPerformance dataset, which is commonly distributed with 1,000 student records and 8 columns. The fields cover gender, race/ethnicity, parental level of education, lunch type, test preparation course, and mathematics, reading, and writing scores. Confirm the row and column counts against the local CSV if a different version of the dataset was used.

Field group	Description
Student/background characteristics	Gender, race/ethnicity, parental education, lunch type
Preparation	Test preparation course
Academic outcomes	Math score, reading score, writing score

3. Tools and Libraries

Python was used in Jupyter Notebook. Pandas supported data loading, inspection, grouping, and descriptive statistics; NumPy was imported for numerical work; Matplotlib and Seaborn were used to create charts.

4. Data Quality Checks

The notebook checks for missing values and duplicate rows, then creates a working copy of the data and removes duplicate records if present. The final missing-value and duplicate counts should be read from the notebook output. No values should be imputed or removed unless the checks indicate a reason to do so.

5. Analysis and Visualizations

Average student scores: Compares the mean mathematics, reading, and writing scores.

Math score distribution: Shows the frequency distribution of mathematics scores and an optional density curve.

Math scores by gender: Uses a box plot to compare score distributions across gender groups.

Test preparation analysis: Compares mean scores for students who completed preparation and those who did not.

Score correlation heatmap: Displays pairwise correlations among the three exam scores.

6. How to Interpret Results

The notebook's printed results provide the actual averages and group comparisons. A positive correlation means that higher values in one score tend to occur alongside higher values in another. Correlation does not show that one subject score causes another. Group differences are descriptive and may reflect other factors not tested here.

7. Key Findings

Insert the exact numerical findings from the completed notebook before final submission. Recommended items include the average score for each subject, which subject has the highest and lowest average, the score differences by test preparation status, the strongest pairwise correlation, and any visible distribution patterns. This report intentionally does not invent numerical results that have not been provided in the conversation.

8. Conclusion

This project demonstrates a standard exploratory data analysis workflow: load and inspect data, assess data quality, summarize numeric variables, visualize distributions and group comparisons, and examine correlations. The analysis helps develop analytical thinking and provides a structured overview of student performance patterns.

9. Project Files

Recommended deliverables: EDA_Student_Performance_Project.ipynb, StudentsPerformance.csv, average_student_scores.png, math_score_distribution.png, math_scores_by_gender.png, test_preparation_analysis.png, score_correlation_heatmap.png, README.md, and this PDF report.

Submission note: verify the findings section against your actual notebook output and make sure all chart files and the dataset are included in the final project folder.